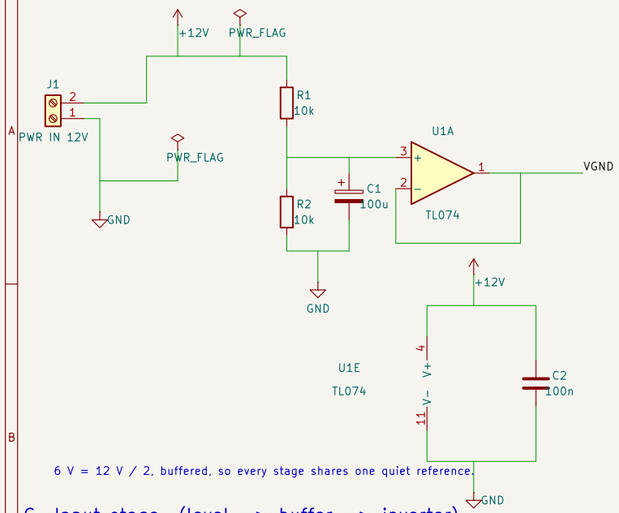
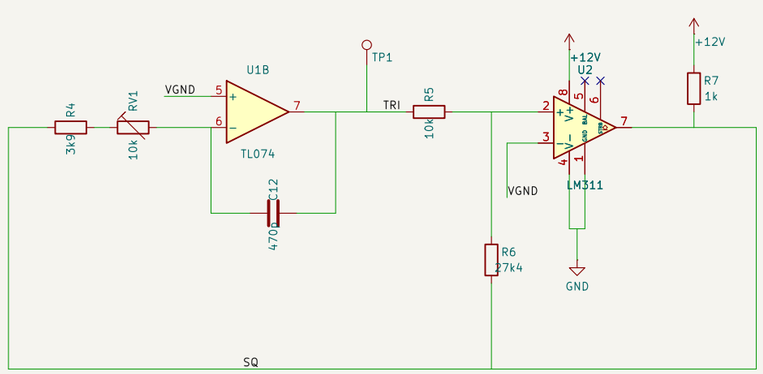
A Power in and 6 V virtual ground



6 V = 12 V / 2, buffered, so every stage shares one quiet reference.

B Triangle carrier ~250 kHz (LM311 Schmitt + TL074 integrator)



f = Vsw / (8 R C): R4+RV1 ~ 5k6 with C12 470p gives 250 kHz; R5/R6 set the +/−2 V window about 6 V



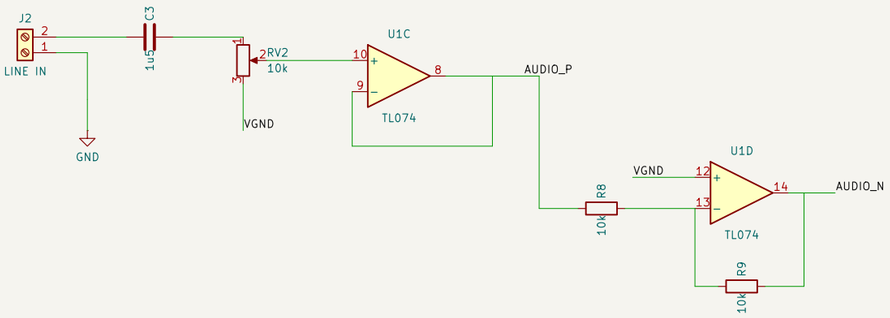
H Indicator and bring-up



Test points TP1..TP5 are solder lugs on TRI, PWM_A, both switching nodes and the filtered output.

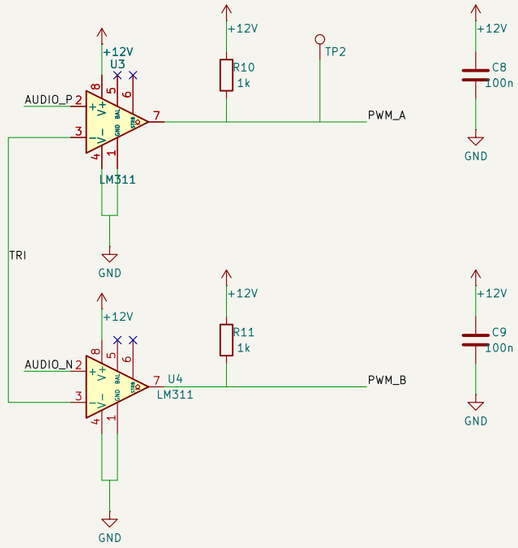
Bring up with NO MOSFETs fitted: check 12 V and 6 V, then the triangle, then both PWM outputs, then the HIP4082 dead time on a scope. Only then fit Q1..Q4, on a current-limited supply into a dummy load.

C Input stage (level -> buffer -> inverter)

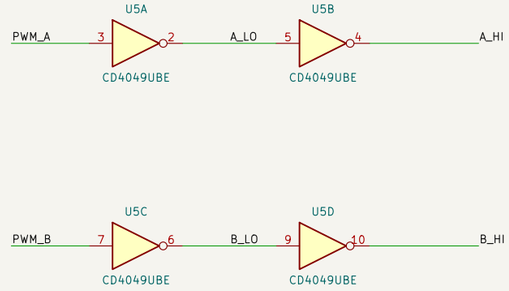


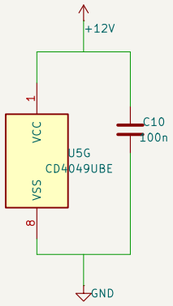
AUDIO_P and AUDIO_N are equal and opposite about 6 V — that is what swings the bridge both ways.

D PWM comparators (audio vs triangle)

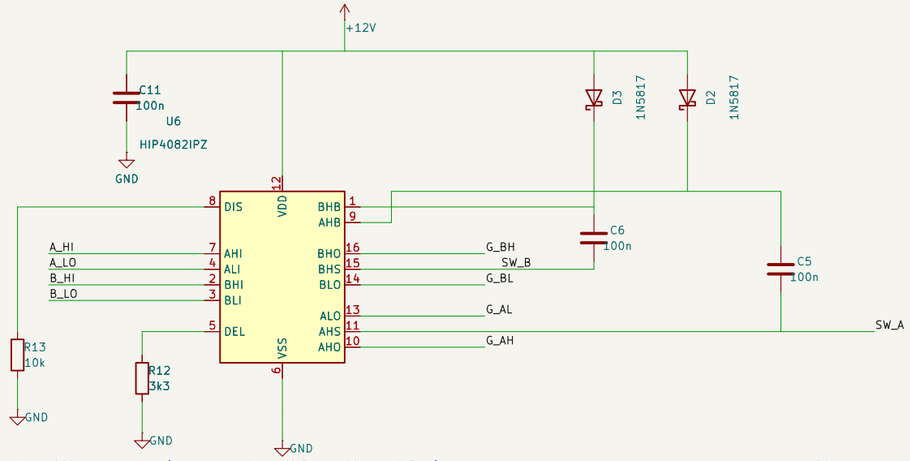


E1 Complement generation (CD4049)





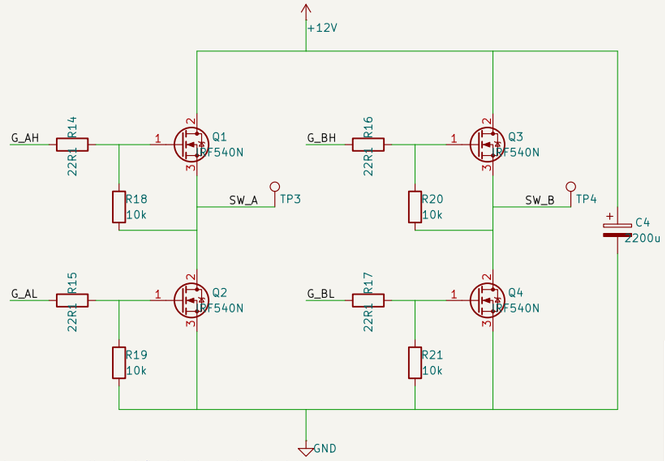
E2 Gate driver HIP4082IPZ



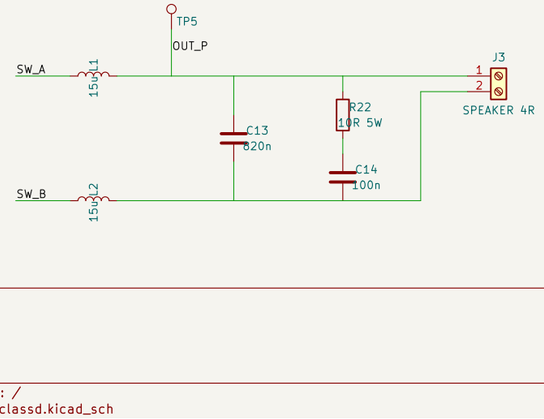
R12 = 3k3 on DEL gives ~200 ns dead time (datasheet: 10k -> 0.5 us, 100k -> 4.5 us).

DIS is active high, so R13 holds it down.

F Output bridge 4 x IRF540N



G Output filter + Zobel



No output DC blocking cap and no speaker ground — the load floats between the two legs.

L1/L2 are hand-wound on shop toroids: the stocked radial chokes saturate below the 3 A peak.

Sheet: /	
File: classd.kicad_sch	
Title: Class D amplifier — mono BTL, 12 V, 4 ohm	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 1/1